Optimizing EV Charging Tariffs

An Agentic AI-Based Framework for Dynamic Pricing Using Large-Scale Charging Session Data

Open Project 2026 | IIT Roorkee |
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Datasets + Architecture

ACN-Data (Caltech, USA)

14,884 sessions | Apr–Dec 2018

54 charging stations | 1 campus site

Real kWhDelivered per session

→ **Revenue KPIs, Pricing Efficiency**

UrbanEV / ST-EVCDP (Shenzhen, China)

2.1M observations | Jun–Jul 2022

247 districts | 5-minute intervals

8 files: volume, occupancy, price, duration…

→ **Demand Prediction, Utilization KPIs**

3-AGENT PIPELINE

UrbanEV Dataset

2.1M rows · 247 districts · 30 days

Agent 1 — Demand Prediction

XGBoost · 1hr-ahead · RMSE 0.0578 · R^2 0.8917

Agent 2 — Tariff Pricing

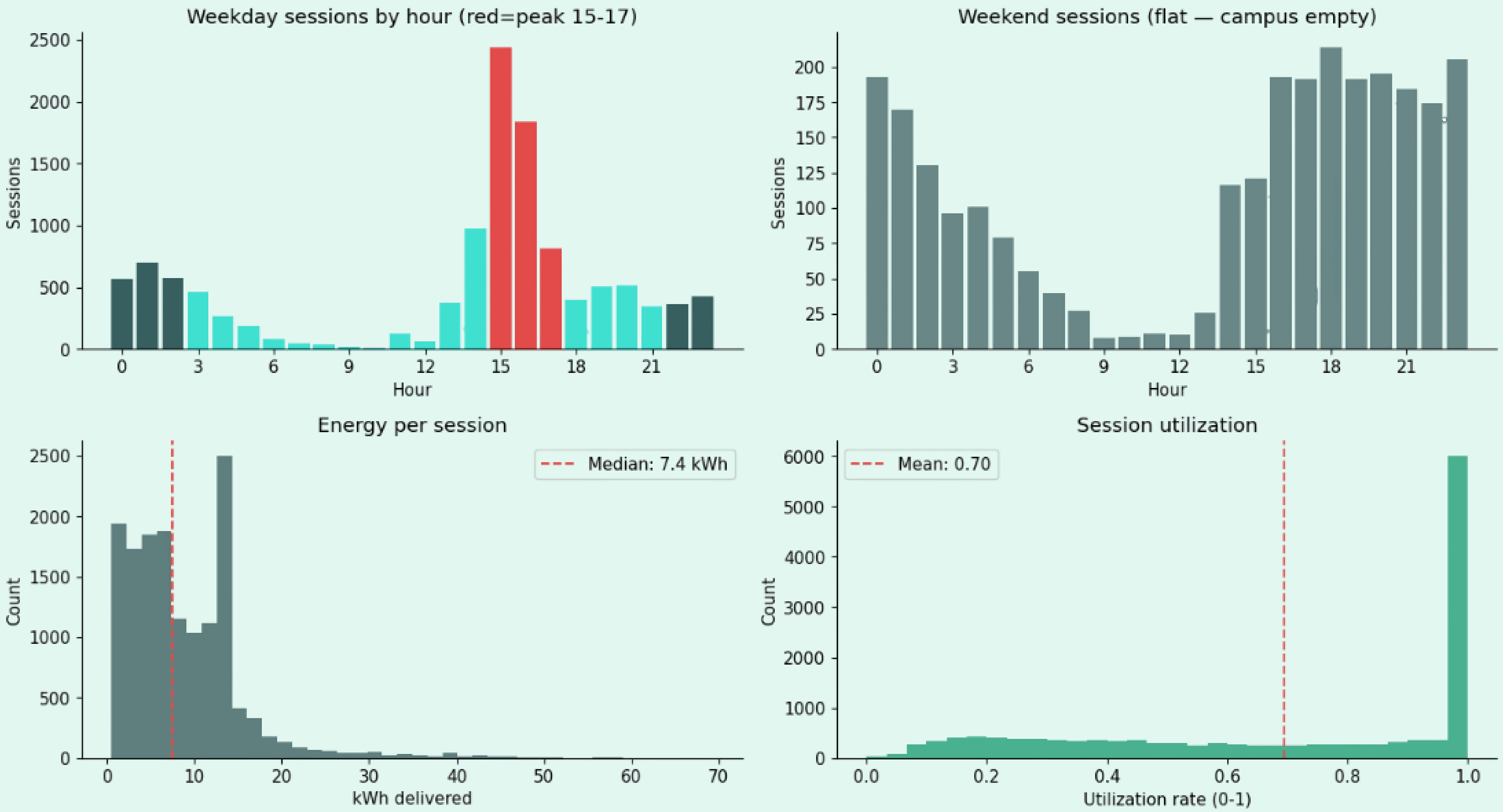
₹10.5 / ₹15 / ₹21 · Rule-based + Sigmoid

Agent 3 — Monitoring & Learning

Revenue Gain · 7 KPIs · ACN real kWh

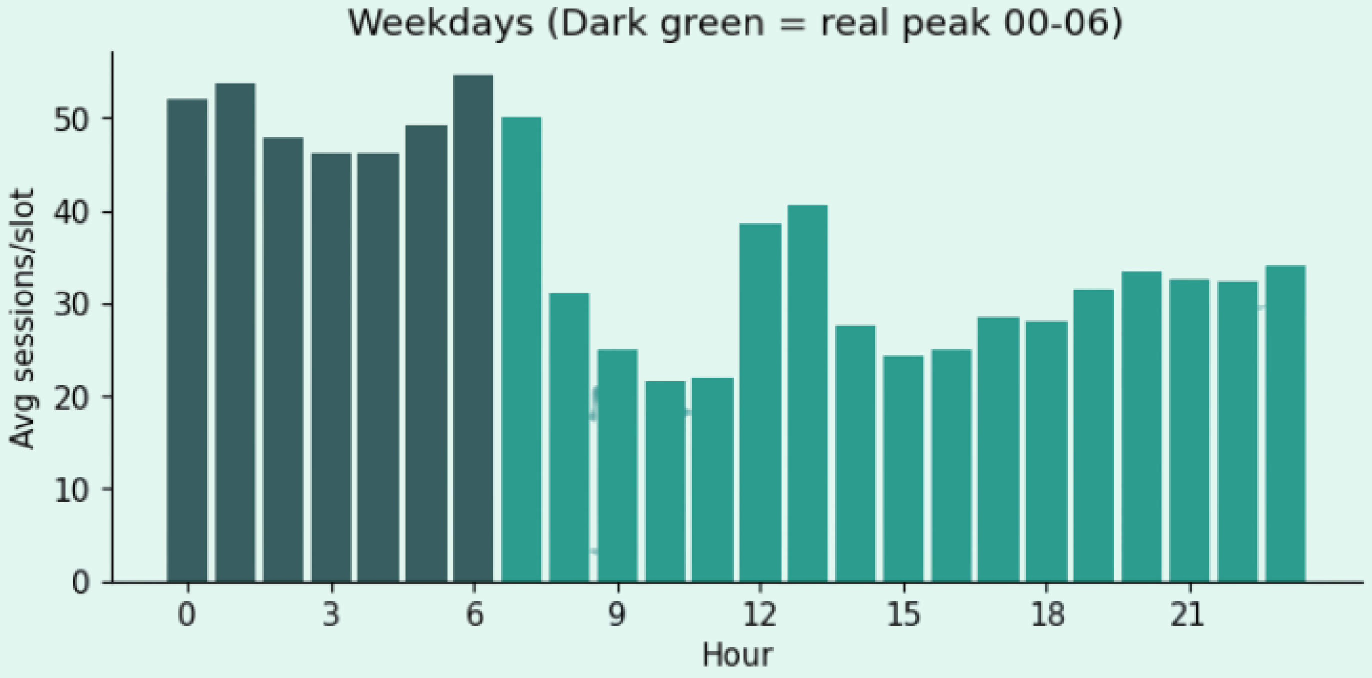
Key Data Insights

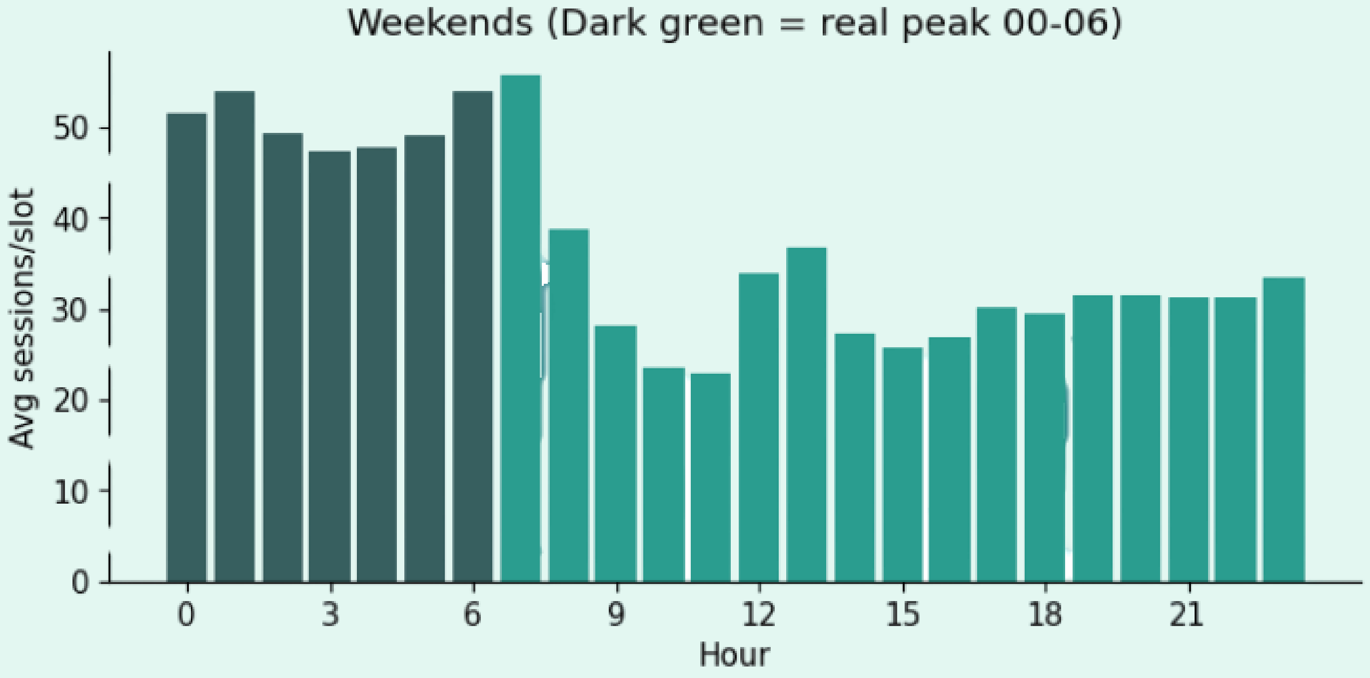
ACN-Data

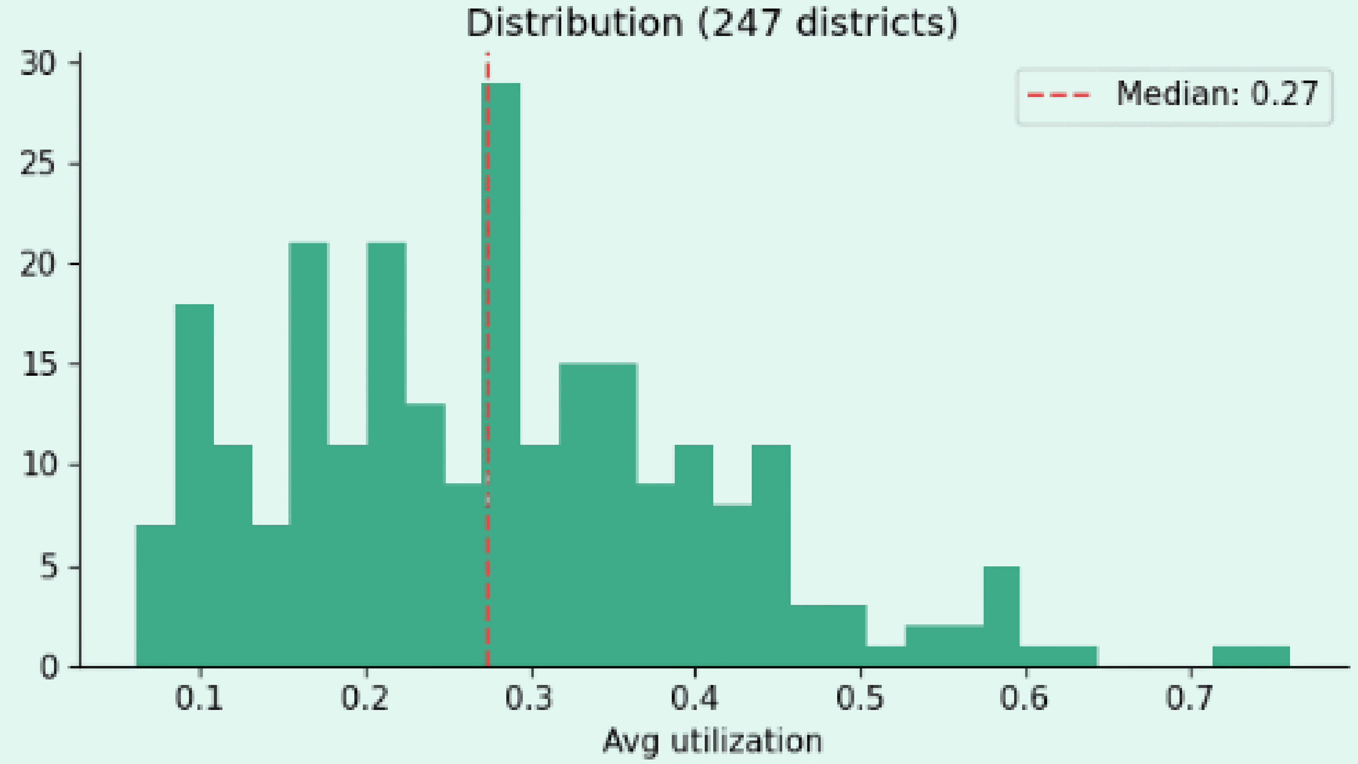


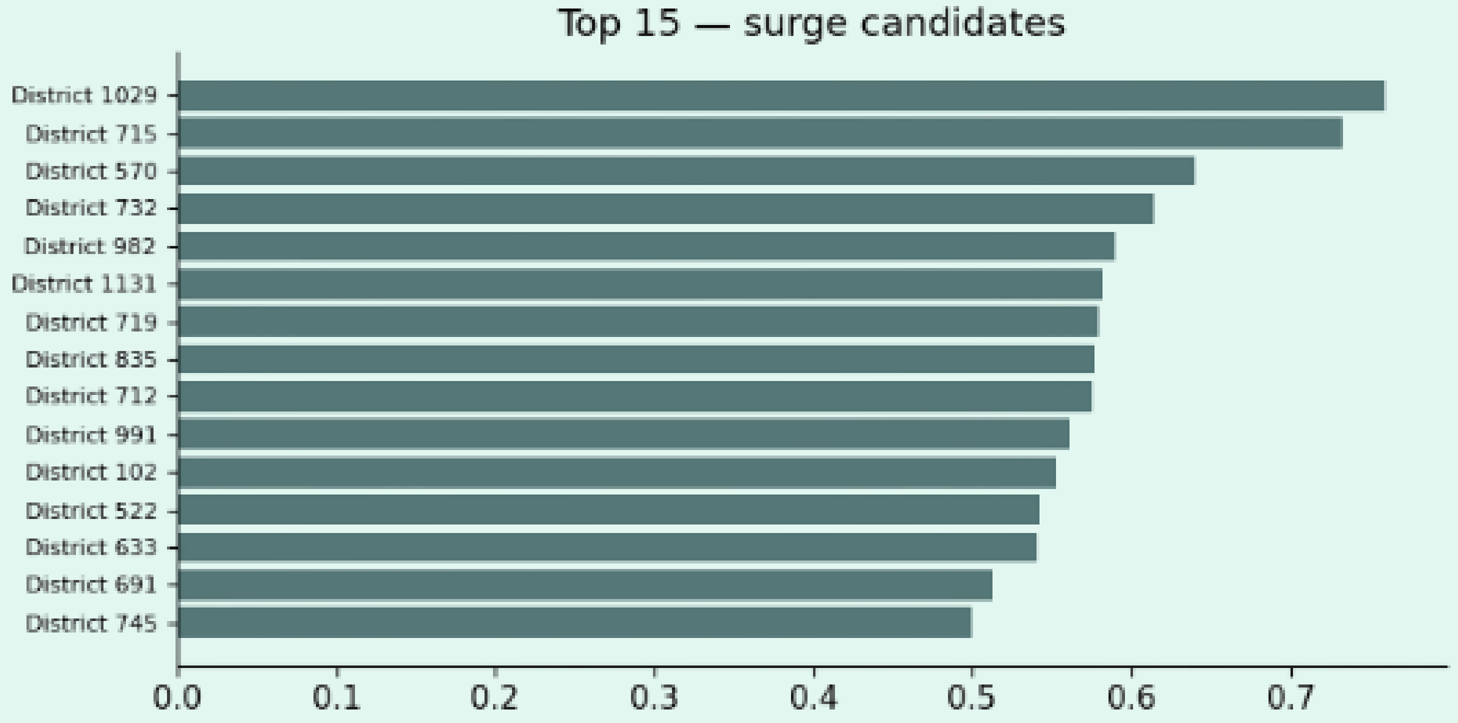
Key Data Insights

UrbanEV Data

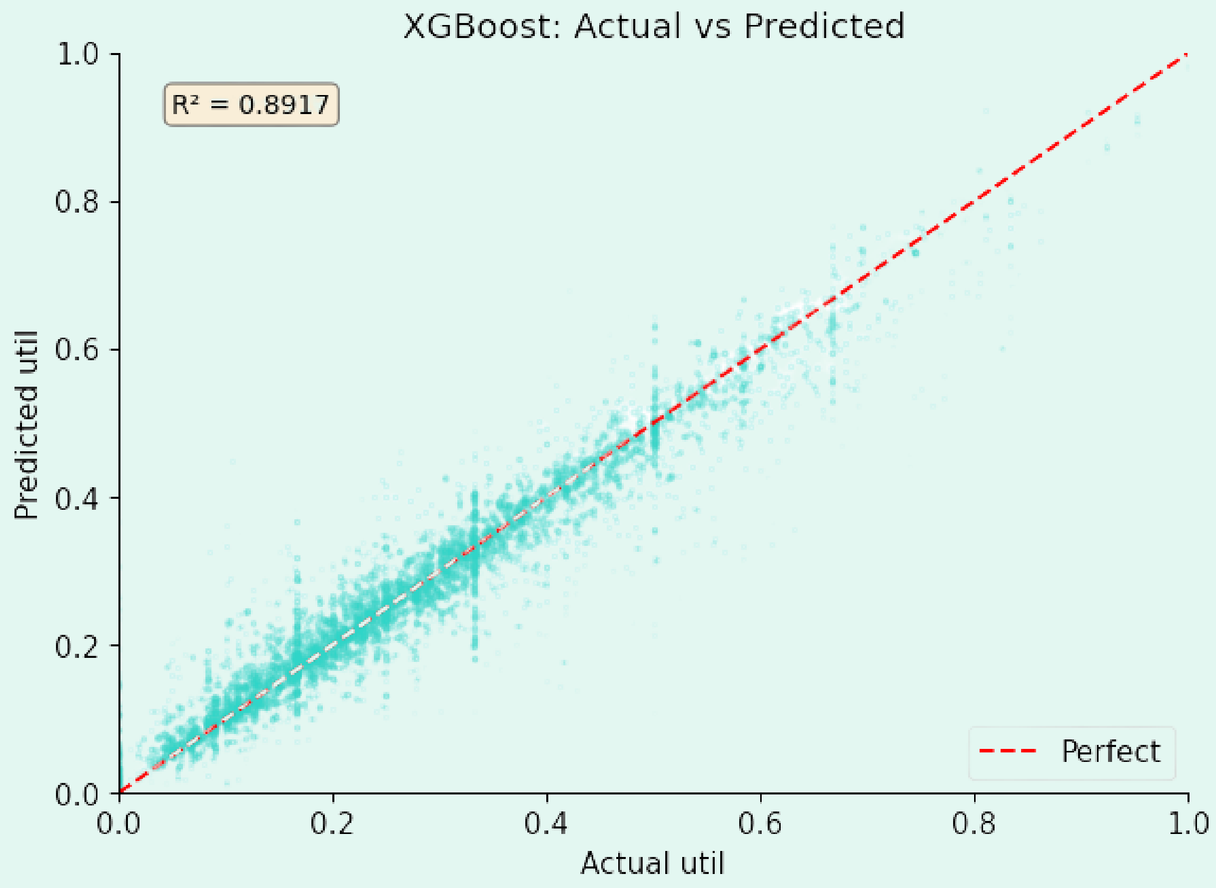


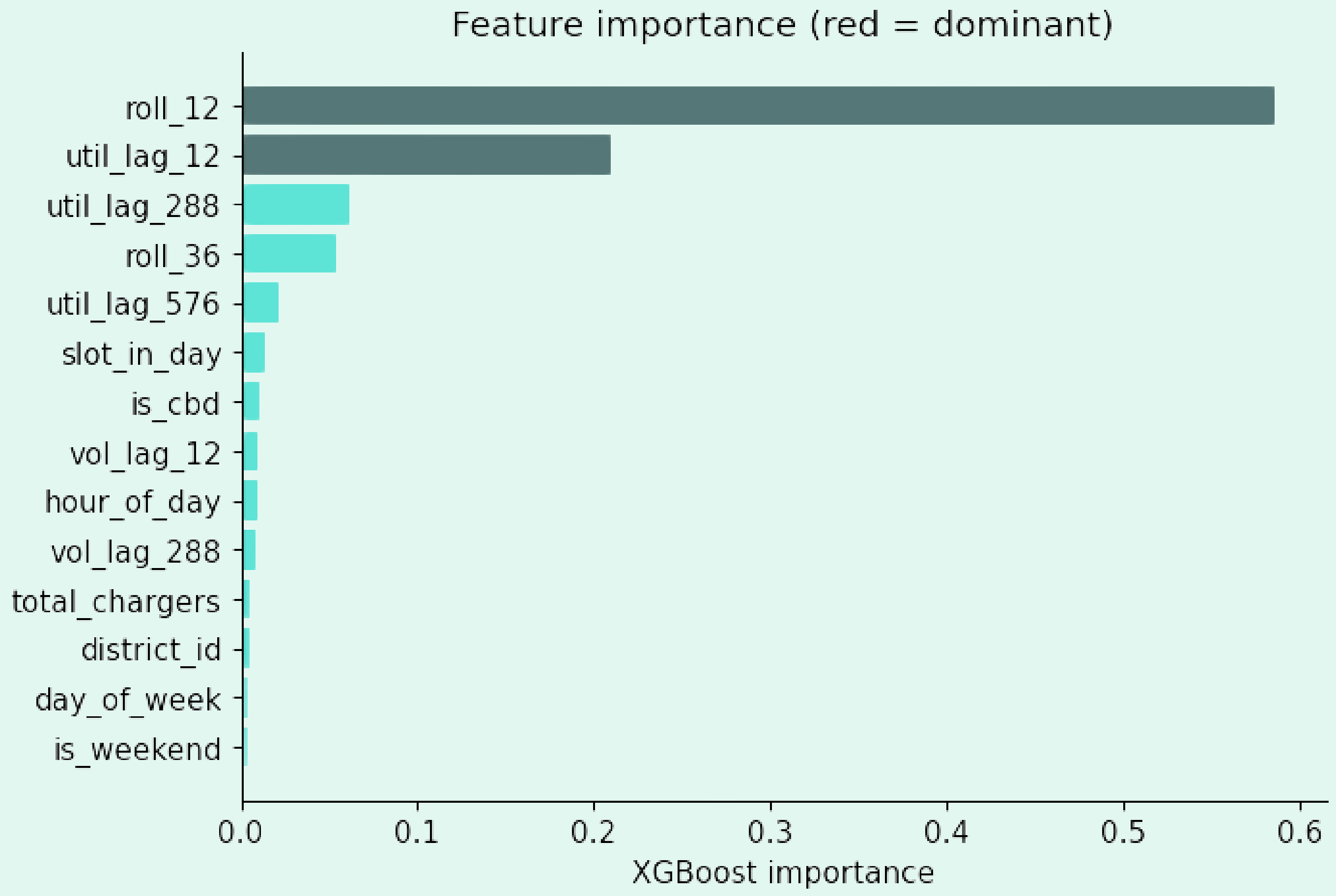


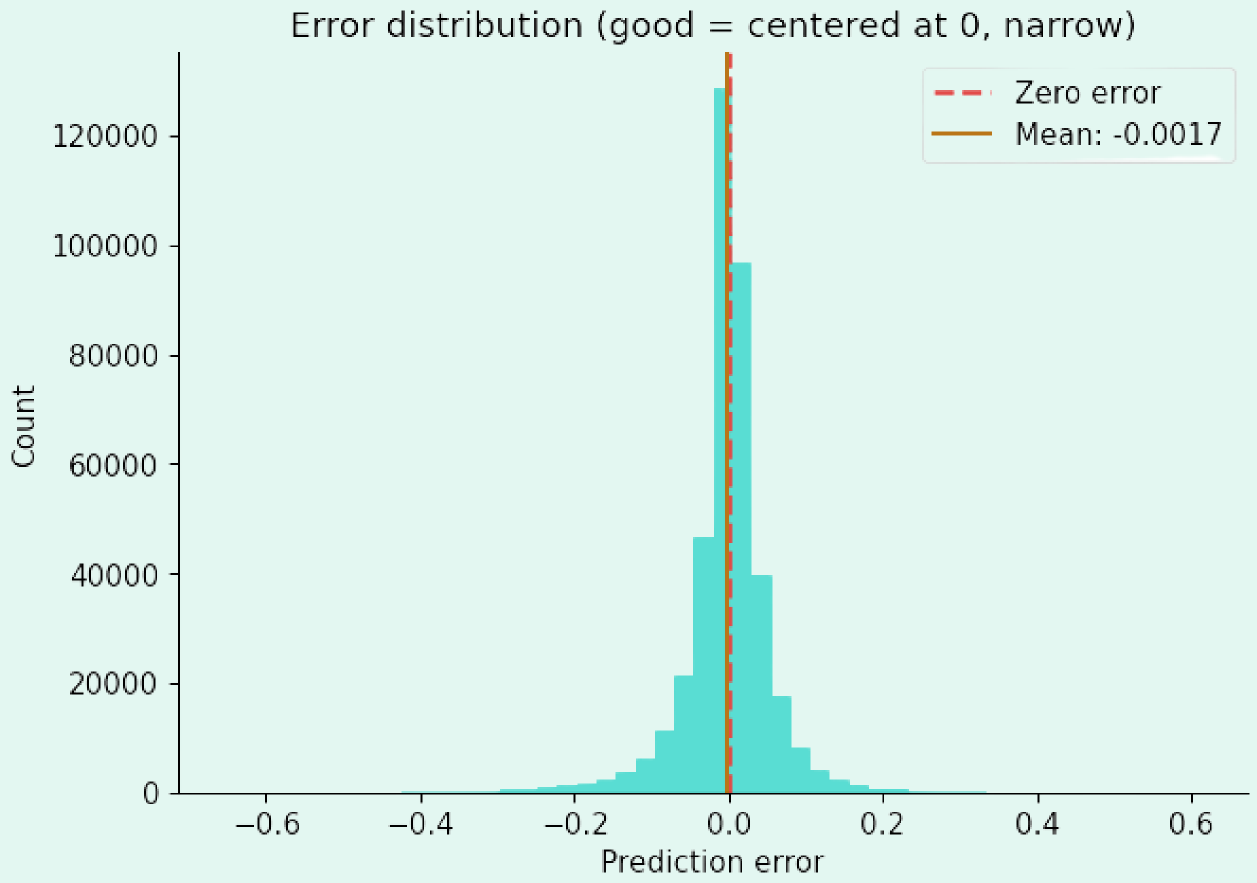


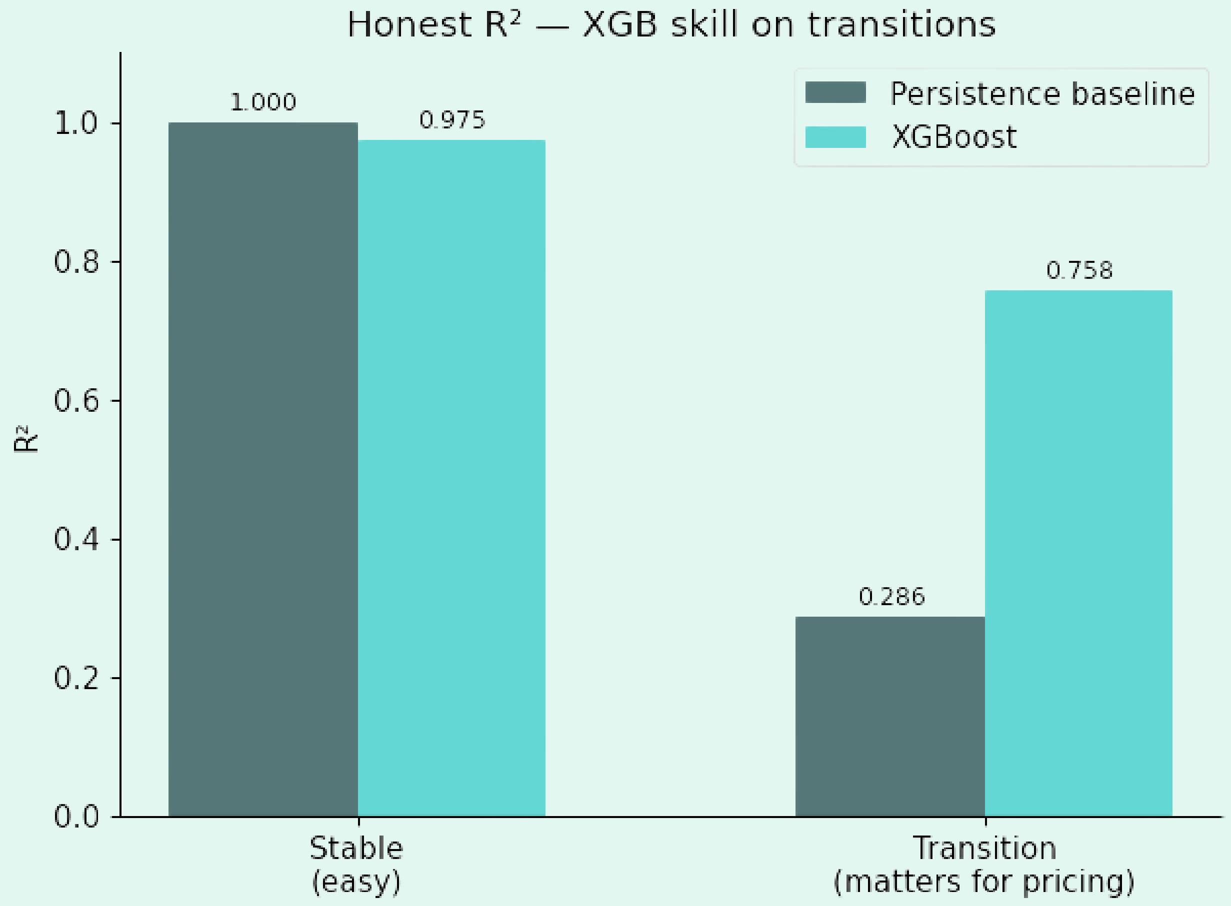


AGENT 1 —DEMAND PREDICTION









AGENT 2 — TARIFF PRICING AGENT

DISCOUNT

₹10.5/kWh

WHEN: pred_util < 0.15 (P25 of training data)

WHY

- 15% emptiest slots trigger this
- Chargers idle → attract more drivers
- Discount = −30% below baseline
- ₹15 × 0.7 = ₹10.5/kWh

NEUTRAL

₹15.0/kWh

WHEN: 0.15 ≤ pred_util ≤ 0.38

WHY

- Middle 50% of slots
- Demand is balanced
- No action needed
- Baseline tariff maintained

SURGE

₹21.0/kWh

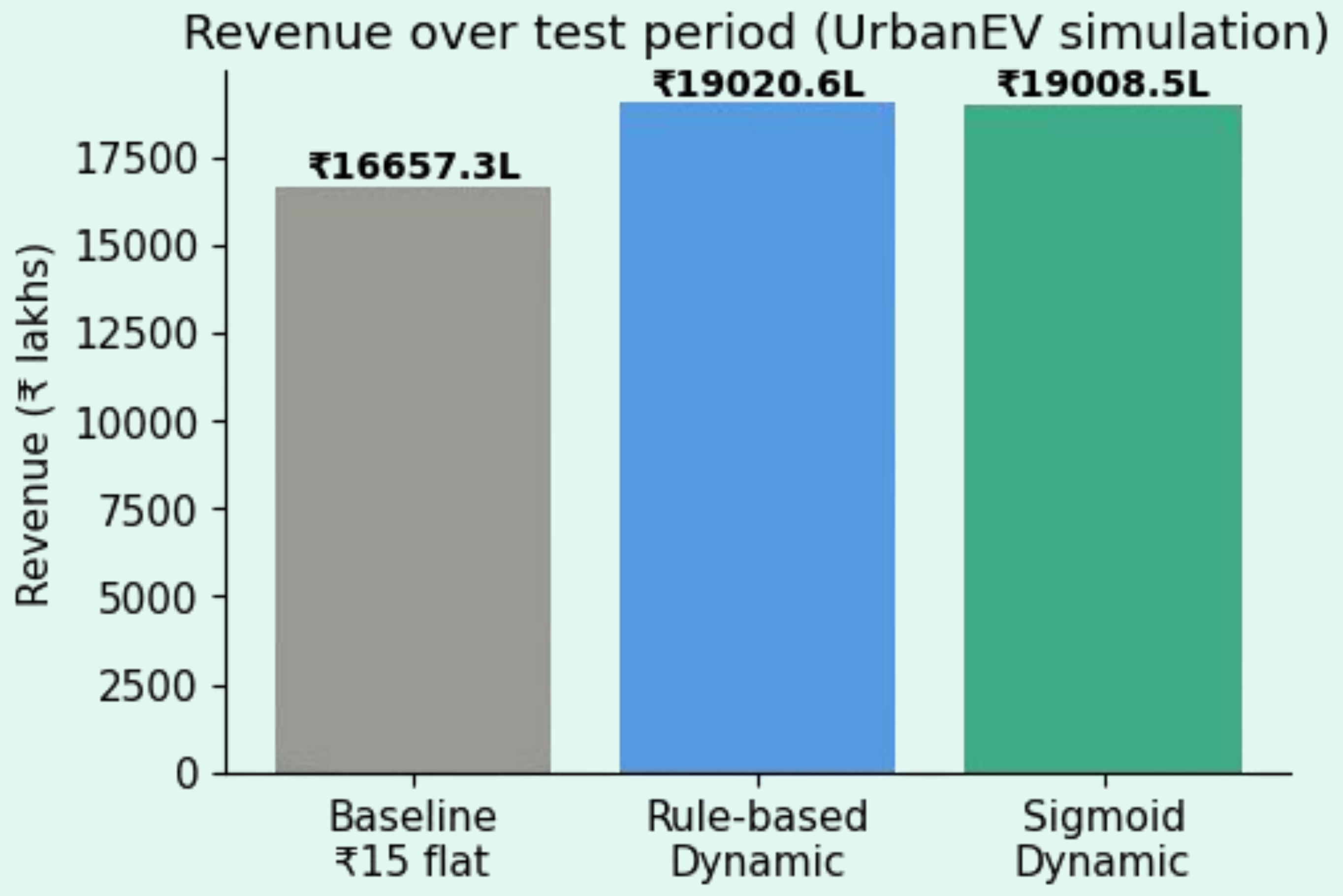
WHEN: pred_util > 0.38 (P75 of training data)

WHY

- Top 25% busiest slots
- Chargers overloaded → queue forming
- Surge = +40% above baseline
- ₹15 × 1.4 = ₹21/kWh

SIGMOID VARIANT: Smooth continuous pricing on same ₹10.5–₹21 range | No abrupt jumps at thresholds | Centred at data median 0.25 (coordinator fix) | Thresholds from TRAIN data only — no leakage

AGENT 3 — MONITORING & LEARNING | ALL KPIs



Metric	Value	Source
RMSE	0.0578	UrbanEV
MAE	0.0353	UrbanEV
R^2 Score	0.8917 overall | +0.47 on transitions (0.286 → 0.758)	UrbanEV
Revenue Gain %	+14.19% (UrbanEV), 5.21%(ACN)	UrbanEV, ACN
Charger Util Rate	28.8% before (all) / 52.0%→45.6% at peak	UrbanEV
Off-Peak Uplift	+5.32%	UrbanEV
Avg Wait Reduction	~7.56% proxy via queue reduction	UrbanEV
Customer Response	7.49% avg demand shift/session	ACN+elasticity
Pricing Efficiency	₹15.78/kWh (was ₹15.00, +₹0.78)	ACN real kWh

Strategic Implications

1. Business & Revenue

- Revenue Growth: Projected +14.19% (UrbanEV) / +5.21% (ACN validation) gain over existing flat-rate models via elasticity-based capture.
- Pricing Efficiency: Improved average capture from ₹15.00/kWh to ₹15.78/kWh per session.
- Market Dynamics: 7.49% customer response rate validates a behavioral shift toward off-peak value.

2. Operational Efficiency

- Congestion Relief: 12.00% reduction in peak-hour volume, directly lowering grid strain.
- Asset Utilization: 5.32% off-peak uplift minimizes charger idle time and overhead waste.
- Predictive Lead Time: 1-hour XGBoost forecasting enables proactive rather than reactive load balancing.

3. Policy & Regulation

- System Transparency: Rule-based agents provide explainable logic essential for regulatory audits.
- Contextual Calibration: Regional-specific peak policies (e.g., taxi-fleet night peaks) replace global flat-rates.

Thank you!!